(i)

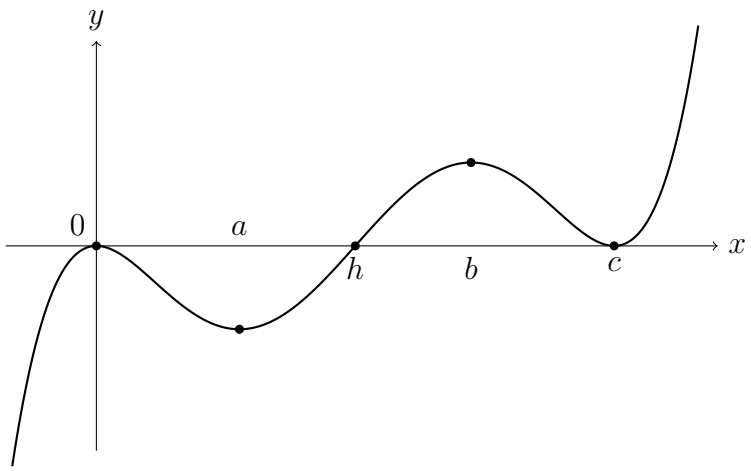
$$f(x) = x(x-a)(x-b)(x-c)$$

Area 0 to a: A

$$F(0) = 0,\ F(a) = -A,\ F(b) = A,\ F(c) = 0$$

Turning points: $x = 0,\ a,\ b,\ c$

x	$-\infty$	0	a	h	b	c	∞
$F(x)$	$-\infty$	0	$-A$	0	A	0	∞



$f(x)$ is monic quartic, so $F(x)$'s leading coefficient is $\frac{1}{5}$

$$F(0) = 0,\ F'(0) = 0, F(c) = 0,\ F'(c) = 0$$

So $0, c$ are repeated roots of $F(x)$

$$F(a) < 0,\ F(b) > 0$$

So there is another root h between a and b

$$F(x) = \frac{1}{5}x^2(x-c)^2(x-h), \quad F(c-x) = \frac{1}{5}x^2(x-c)^2(c-h-x)$$

$$\begin{aligned} F(x) + F(c-x) &= \frac{1}{5}x^2(x-c)^2(x-h) + \frac{1}{5}x^2(x-c)^2(c-h-x) \\ &= \frac{1}{5}x^2(x-c)^2(c-2h) \end{aligned}$$

(ii)

For $0 \leq x \leq c$

$$F(x) \geq -A,\ F(x) = -A \iff x = a,\ F(b) = A$$

$$\therefore F(b) + F(x) \geq 0$$

$$F(b) + F(x) > 0 \quad (x \neq a)$$

$$F(b) + F(c-b) = \frac{1}{5}b^2(c-b)^2(c-2h)$$

$$c - b = a \quad \text{or} \quad F(b) + F(c-b) > 0$$

$$c - b = a \quad \text{or} \quad c > 2h$$

Similarly,

$$F(a) + F(x) \leq 0, \quad F(a) + F(x) < 0 \quad (x \neq b)$$

$$F(a) + F(c-a) = \frac{1}{5}a^2(c-a)^2(c-2h)$$

$$c - a = b \quad \text{or} \quad c < 2h$$

$$c > 2h \text{ and } c < 2h \text{ cannot both hold}$$

$$\therefore c = a + b$$

$$F(b) + F(a) = 0$$

$$\frac{1}{5}b^2(c-b)^2(c-2h) = 0 \implies c = 2h$$

(iii)

$$h = \frac{c}{2}$$

$$F(x) = \frac{1}{5}x^2(x-c)^2\left(x - \frac{c}{2}\right)$$

$$
\begin{aligned}
f(x) &= F'(x) \\
&= \frac{1}{5}x(x-c)(5x^2-5cx+c^2) \\
&= x(x-c)\left(x^2-cx+\frac{c^2}{5}\right) \\
&= \frac{1}{5}(5x^4-10cx^3+6c^2x^2-c^3x)
\end{aligned}
$$

$$
f''(x) = \frac{1}{5}(60x^2-60cx+12c^2) = 12\left(x^2-cx+\frac{c^2}{5}\right)
$$

$$
\because f(x) = x(x-c)\left(x^2-cx+\frac{c^2}{5}\right)
$$

$$
\therefore \text{When } f''(x) = 0 \implies f(x) = 0
$$

The points of inflection on $y = f(x)$ lie on the x-axis

This solution was submitted by **Jon**

Published and refined by **Jon**

Last updated: July 18, 2026